
CHAPTER

8

RELIABILITY AND VALIDITY IN MEASUREMENT

Learning Outcomes

On completion of this chapter, the users will be able to

- *Know about the errors in measurement;*
 - *Understand about the reliability and validity and their measurements;*
 - *Learn about standard error of measurement;*
 - *Construct measurement scale.*
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8.1 Measurement Error

Measurement is at the heart of any scientific discipline. The primary function of measurement is to assign numbers to objects or events according to certain rules (Stevens, 1951: 22). It is the process by which empirical data are organized in some systematic relationship to the concept being studied. Bryman (2004: 66) identifies three main reasons for the preoccupation with measurement in quantitative research:

- Measurement allows us to delineate fine differences between

people in terms of the characteristics in question.

- Measurement gives us a consistent device or yardstick for making such distinctions
- Measurement provides the basis for more precise estimates of the degree of relationship between concepts (e.g. through correlation analysis).

Measurement theory contends that however precise our instruments for measuring and however careful our efforts of observations are, there will always be some errors introduced into our measurement. This is what we refer to as **measurement error**. In fact, the central formulation of measurement theory states that an observed measure (score) is equal to the true score plus some error (above or below the true score), which necessarily occurs in the process of observing the phenomenon. Thus if X is an observed score, T a true score and e a measurement error, then the relationship between X , T and e for the i -th individual can be expressed as:

$$X_i = T_i + e_i \quad (a)$$

Thus for a specified measurement i , the measurement error e_i is the difference between X_i and T_i :

$$e_i = X_i - T_i \quad (b)$$

Whenever there is a measurement of an attribute, three kinds of error are likely to be generated in the measurement process. These are (i) random error, (b) systematic error, and (c) situational error. **Random error** appears as an unexplained variation. It occurs in a random fashion. A random error exhibits no systematic tendency to be either negative or positive. If the errors are truly random, they will cancel each other and the overall average of these errors will be zero. That is

$$\sum e_i = \sum (X_i - T_i) = 0 \quad (c)$$

Sometimes people do not know their exact age. While reporting, some people will overstate their age, and some will understate. This will produce random error.

Inaccuracy of measurements that occurs repeatedly and for the same reasons is labeled **systematic error**. Observations, which consistently overestimate or underestimate the true value, are subject to systematic error. Systematic errors frequently occur as a result of improper research procedures such as

- Faulty conceptualization of the research problem;

- A poorly designed research instrument;
- The researcher's bias; and
- Incorrect operationalization of variables.

Unlike random error, the sum of the systematic errors does not add to zero. In the measurement of contraceptive use, for example, if people consistently report higher use, because they believe this would be appreciated by the interviewer or the program managers, errors would take place systematically giving a false high rate of contraceptive use. Likewise, a defective weighing machine or any other defective measuring device is likely to produce systematic error in the measuring process.

Situational errors are ones that occur neither at random nor systematically, but occur as a result of particular personal characteristics of the individual(s) or other special circumstances that can influence the measurement process. Fatigue, for example, of the respondents or an unusual environmental distraction may generate situational error.

Social researchers primarily employ two criteria in evaluating the quality of their measures. These are (i) reliability, and (ii) validity. These two criteria seek to answer: how consistent or dependable (which refers to reliability) and accurate (which refers to validity) are the information obtained? We discuss these concepts in turn in the following sections.

8.2 Reliability and its Measurement

Put simply, reliability is the degree to which a measuring procedure produces similar, in other words consistent outcomes, when it is repeated under similar conditions. A tailor measuring fabric with a measuring tape obtains a value of the length of the fabric as 20 inches. If the tailor takes repeated measures of the fabric and each time comes up with the same length, it is said that the tape measure is reliable. The essential conditions in such measurement process are that repeated administration of the same instrument must be made under essentially similar conditions and the object will not be changed over time. Reliable instruments are robust; they work well at different times under different conditions. However, if the object being measured does change in value, the reliability measure will indicate that change.

How can a measuring instrument be unreliable? If your weight remains constant at 60 kilograms but repeated measurements on your bathroom scale show your weight to fluctuate, the lack of reliability may be due to a weak spring inside the scale. In the case of a social research, unreliability

in questionnaire as a measuring instrument, might arise in the measurement of any concept due to such things as question or answer categories being so ambiguous that the respondent is unsure how he or she should answer. In the given circumstances, the respondent fails to answer consistently. This leads to unreliability of the measuring instrument (here the questionnaire). The credibility and testability of any theory thus depend primarily upon adequate measurement tools or devices in social research.

The reliability of a measure can be described in terms of three interrelated concepts:

- Stability;
- Equivalence; and
- Internal consistency.

A measure is said to be **stable** if you can secure consistent results with repeated measurements of the same individuals or objects with the same instruments. An observational procedure or measuring device is stable if it gives the same reading on an individual or object when repeated one or more times.

The second perspective on reliability, i.e. **equivalence**, considers how much error may be introduced by different investigators (in observation) or different samples of items being studied (in questioning or scales). Thus, while stability is concerned with personal and situational fluctuations from one time to another, equivalence is concerned with variations at one point in time among observers and samples of items. A good way to assess the equivalence of measurement by different observers is to compare their scoring or evaluating of the same event. In 1993 IDD prevalence survey in Bangladesh, for example, 8 physicians measured the goiter grade of a single person at a single point of time to assess the extent of error incurred in the measurement due to the physicians.

The third approach to reliability, i.e. the **internal consistency**, uses only one administration of an instrument or test to assess the consistency of homogeneity among the items.

It must always be kept in mind that the reliability refers to a specific measuring instrument applied to a specific population under specific condition. To assess the reliability of any test based on at least interval data, we can use, in general, the Pearson's product moment correlation coefficient r , and the resulting correlation will be our estimate of the reliability coefficient. The coefficient varies between '0' (no correlation and therefore no internal consistency) to '1' (perfect correlation and

therefore complete internal consistency). It is usually expected that a coefficient of 0.8 and above implies an acceptable level of reliability. The correlation must be significant statistically in order to be used to establish reliability and the strength of the correlation provides information regarding the dependability of the measures involved.

In estimating reliability, one can make use of different sources of variation in measurement values. The simplest approach is the so called variance approach, which makes use of the foregoing equation (a), viz. $X_i = T_i + e_i$.

The set of observed score has a total variance V_0 (say). Likewise, the true scores (T_i) and the errors of measurement e_i will have variances V_t and V_e respectively. The total variance is the sum of the true variance and the error variance, that is

$$V_0 = V_t + V_e \quad \dots (d)$$

The ratio of the true variance to the total variance is the so called **reliability coefficient**. That is

$$\text{Reliability coefficient } (r_c) = \frac{V_t}{V_0} \quad \dots (e)$$

Since V_t is less than or equal to V_0 , the reliability coefficient can take on a maximum value of 1. When this is so, we say that the observed scores are exactly the same as the true scores and in that instance a perfect reliability has been assumed to be achieved.

An alternative formula for the reliability coefficient can be obtained by dividing both sides of the expression (d) by V_0 . Thus

$$\frac{V_0}{V_0} = \frac{V_t}{V_0} + \frac{V_e}{V_0} \quad \text{or} \quad 1 = r_c + \frac{V_e}{V_0}$$

After some rearrangements

$$r_c = \frac{V_0 - V_e}{V_0} \quad \dots (f)$$

There are a number of other popular techniques available for measuring the reliability. These are

- Test-retest method
- Parallel-forms method
- Split-half method,

approximately equal in difficulty for this purpose. A correlation coefficient can then be computed and its statistical significance be tested.

We illustrate the computation and interpretation of parallel form reliability by another example due to Sprinthall (1997: 429).

Example 8.2 The Sound Recognition Test is a test for a condition known as auditory agnosia, or a person's ability to recognize familiar environmental sounds, such as a bell, a whistle, or crowd's sounds. There are two forms of the test, A and B, with a total of 13 items per test. Scoring is based on allowing up to 3 points per item, making 39, the highest possible score. A group of normal, five-year old children was selected and given form A. Then the next day they were given form B. The accompanying table shows the data and the scheme of calculating the reliability coefficient.

	Form A		Form B		
Subject	X	y	x^2	y^2	xy
1	30	29	900	841	870
2	25	32	625	484	550
3	35	31	1225	961	1085
4	22	25	484	625	550
5	38	36	1444	1296	1368
6	31	31	961	961	961
7	29	30	841	900	870
8	28	30	784	900	840
9	32	34	1024	1156	1088
10	30	32	900	1024	960
Total	300	300	9188	9148	9142

The product moment correlation coefficient is calculated as follows:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}}$$

$$= \frac{(10)(9142) - (300)(300)}{\sqrt{[(10)(9188) - (300)^2][(10)(9148) - (300)^2]}} = 0.851$$

With 8 df, the product moment correlation coefficient r is significant at 5% level (a table value of .632 is required). Thus the parallel form reliability is established at .851, which again is an acceptable value for a test of this type.

Bangladesh Bureau of Statistics (BBS) has made use of both the test-retest and parallel-forms methods to assess the reliability of reporting of the births and deaths in their vital registration system. Under the test-retest method, data were collected under system-1 by the registrar on regular basis. This system was referred to as 'test'. Under system-2, retrospective data for the last three months were collected by the supervisors, which they called retest (BBS, 2004: 111).

We now turn to discuss internal consistency techniques. These techniques are referred to as internal consistency technique because what is being assessed in this case is the consistency that occurs within the test rather than between two administrations of the same test or separate administration of the alternate forms. The methods to be discussed are

- (i) Split-half method
- (ii) Kuder-Richrdson formula
- (iii) ANOVA

8.2.3 Split-half Method

The parallel-forms method in many instances is expensive and frequently difficult to construct. Therefore, a less direct method of assessing the effects of different samples of items is the so called **split-half** method. Only one test is administered to compute reliability coefficient in this technique. The whole set of test-items is divided into two equal halves. Then a test can be administered and separate scores assigned to every individual on two arbitrarily selected halves of that test can be obtained. For example, an individual may be given one score on the odd-numbered items, and a second score on the even-numbered items. Then the product-moment correlation between the two sets of scores gives the parallel-forms reliability coefficient for a test half as long as the original test.

A notable problem arises in the split-half technique while splitting the test in order to obtain the most nearly comparable halves. In most tests, the first half and the second half would not be comparable owing to differences in nature and difficulty level of items. A procedure that is adequate for most purposes is to find the scores on odd and even numbered items of the test. If the items were originally arranged in an approximate order of difficulty, such a division yields very nearly equivalent half-scores.

Once the two-half scores have been obtained for each individual, they may be correlated by the usual method. It should be noted; however, that correlation actually gives the reliability of only a half test. For example, if

the entire test consists of 100 items, the correlation is computed between sets of scores each of which is based on only 50 items. In both test-retest and parallel-forms reliability, on the other hand, each score is based on the full number of items in the test. Assuming that the two halves are equivalent, the reliability of the full test (r_{tt}) can be estimated by means of the Spearman-Brown prophecy formula as given below:

$$r_{tt} = \frac{2r_{hh}}{1 + r_{hh}} \quad (d)$$

where r_{hh} is the correlation of the half test. As an example, if the correlation of the total scores on the odd-numbered items with total scores of the even-numbered items is 0.80, the estimated reliability of the whole test is

$$r_{tt} = \frac{2(80)}{1 + 80} = 0.89$$

The Spearman-Brown prophecy formula stated in (d) above is a particular case of a more general formula of the following type:

$$r_{tt} = \frac{kr_0}{1 + (k-1)r_0} \quad (e)$$

in which k is the factor by which the test is to be lengthened or shortened with respect to the original test whose reliability coefficient is r_0 .

The formula (e) is of particular significance. It gives an estimate of the effect of lengthening or shortening a test on its reliability coefficient. Suppose we have an n_1 -item test and we know its reliability, which is r_0 . By the use of this formula, we can predict what its reliability would be if n_2 additional similar items were added to the test. Here $k = (n_1 + n_2)/n_1$. Substituting k in (e), the reliability coefficient of the new test can be calculated. Similarly, if we have an m_1 -item test of known reliability and we wish to reduce it to a test of m_2 -item ($m_2 < m_1$), Spearman-Brown formula may be employed with $k = (m_1 - m_2)/m_1$ to estimate the reliability of the shortened test.

In addition, the formula is useful when we want to determine how many items will be needed to achieve a given level of reliability. For instance, if r_{tt} were set at 0.9, we can determine how many items would be needed inter-correlating 0.5 to achieve this desired level of reliability. This can be obtained from the following formula.

$$k = \frac{r_{tt}(1 - r_0)}{r_0(1 - r_{tt})} \quad (f)$$

The formula can easily be obtained from a sample rearrangement of (e). Setting $r_n=0.90$, and $r_0=0.50$, we find

$$k = \frac{.90(1-.50)}{.50(1-.90)} = 9$$

This demonstrates the main justification for including a number of items in a test (or scale) that the reliability can thereby be increased to a satisfactory level. The formula further shows that the number of items needed to reach a given level of reliability depends on the homogeneity of the items, that is on the inter-correlations between them.

Example 8.3: Suppose we have a 20-item test with a reliability coefficient 0.60. Estimate what the reliability of this test would be if 80 similar items were added to make it a 100-item test.

Solution: In this instance $k = (n_1+n_2)/n_1 = (20+80)/20=5$ and $r_0=0.60$. Hence using (e)

$$r_n = \frac{5(.60)}{1 + (5-1)(.60)} = 0.88$$

Example 8.4: Suppose we have a 110-item test, the length of which is reduced to 55 items. The reliability coefficient of the original test is 0.80. What would be the reliability of the shortened test?

Solution: Here $k = (m_1-m_2)/m_1 = (110-55)/110=0.50$ and $r_0=0.80$. Hence

$$r_n = \frac{.50(.80)}{1 + (.5-1)(.80)} = 0.67$$

An alternative method for finding split-half reliability was developed by Rulon (1939). It requires only the variance of the differences between each person's scores and the two half-tests (s_e^2) and the variance of total scores (s_t^2); these two values are substituted in the following formula, which yields the reliability of the whole test directly:

$$r_n = 1 - \frac{s_e^2}{s_t^2} \quad (g)$$

Thus for a test with standard deviation of 6 and standard error of measurement 3, the Rulon's method gives a reliability coefficient of

$$r_{tt} = 1 - \frac{3^2}{6^2} = 0.75$$

Example 8.5: A sample of ten students, all of the same age, was selected and given a Math test to assess strengths and weaknesses in a number of math and math-related areas. Use split-half method with a Spearman-Brown correction for the desired assessment. The scaled scores for the odd and even items are shown in the accompanying table.

	Odd Even				
Subject	x	y	x^2	y^2	xy
1	12	12	144	144	144
2	10	10	100	100	100
3	7	8	49	64	56
4	9	9	81	81	81
5	11	10	121	100	110
6	9	10	81	100	90
7	10	11	100	121	110
8	9	9	81	81	81
9	10	10	100	100	100
10	10	10	100	100	100
Total	97	99	957	991	982

The computed correlation coefficient is

$$\begin{aligned}
 r &= \frac{n \sum xy - \sum x \sum y}{\sqrt{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}} \\
 &= \frac{(10)(982) - (97)(99)}{\sqrt{[(10)(957) - (97)^2][(10)(991) - (99)^2]}} = 0.883
 \end{aligned}$$

With 8 df, the r is significant at 1% level (for r to be significant a table value of 0.765 is required). Since r is significant, we apply the Spearman-Brown formula as follows:

$$r_{tt} = \frac{2r_{hh}}{1 + r_{hh}} = \frac{(2)(.883)}{1 + .883} = \frac{1.766}{1.883} = 0.938$$

The reliability is thus established at 0.938, which is certainly a lofty value and indicates a high degree of reliability.

8.2.4 Kuder-Richardson (K-R) Formula

A fourth method for finding reliability, also utilizing a single administration of a single form, is based on the consistency of responses to all items in the test. Such a formula is developed by Kuder and Richardson (1937). Rather than requiring two-half-scores, this technique is based on an examination of performance on each item. Of the various formulas developed in the original article, the most widely applicable, commonly known as **Kuder-Richardson formula 20** is the following:

$$r_{tt} = \frac{n}{n-1} \left[1 - \frac{\sum p_i q_i}{s^2} \right] \quad (h)$$

where n is the number of items on the test, s^2 is the variance of the total test scores, and p_i the proportion of persons getting i -th item correct. The product of p and q is computed for each item and these products are then summed over all n items to give $\sum p_i q_i$.

Under the assumption that all items are of equal difficulty, i.e. p_i are the same for items, Kuder and Richardson used an alternative formula for the above expression in the following form:

$$r_{tt} = \frac{n}{n-1} \left[1 - \frac{\bar{x}(n-\bar{x})}{ns^2} \right] = \frac{ns^2 - \bar{x}(n-\bar{x})}{s^2(n-1)} \quad (i)$$

where $\bar{x}$, which equals np , is the mean of the total scores. This formula is known as **Kuder-Richardson formula 21**. The coefficient computed from formula (i) is more conservative estimate of reliability than that obtained from formula (h). Formula (i), which was derived from (h) by making p_i the same for all items, is simpler from a computational standpoint, because the n products $p_i q_i$ do not have to be calculated.

To illustrate the application of formula (i), assume that the mean of a test containing 70 items is 50 and the variance is 100. Then on applying (h), the estimated reliability of this test is

$$r_{tt} = \frac{70}{70-1} \left[1 - \frac{50(70-50)}{70(100)} \right] = 0.87$$

The formula K-R 21 is quick and easy to calculate and when used appropriately, it can be extremely useful in determining a test's overall reliability. Unlike the Split-half method, which splits the test just once, the K-R 21 estimates the reliability of a test that has been split into all possible

halves and it automatically corrects for the splits without any need for a Spearman-Brown type adjustment.

To use the $K-R$ 21, the following criteria should be kept in mind:

- The entire test should be aimed at tapping a single domain. If the test is not clearly focused on a single underlying concept, the reliability value will be underestimated
- The test is scored on the basis of each item being either right or wrong.
- All items have about the same degree of difficulty. The formula works best (produces its highest reliability estimate) when the difficulty index is approximately 0.50 for each item.

Now-a-days, most researchers use a test of internal reliability known as Cronbach's alpha (Cronbach, 1951). It essentially calculates the average of all possible split-half reliability coefficients. A computed alpha coefficient will vary between '1' (implying perfect internal reliability) and '0' (implying no internal reliability). A figure 0.80 is typically employed as rule of thumb to denote an acceptable level of internal reliability, though many writers work with a slightly lower figure. The coefficient is defined as

$$\alpha = \frac{n}{n-1} \left[1 - \sum \left(\frac{s_i^2}{s_t^2} \right) \right] = \frac{n}{n-1} \left[\frac{s_t^2 - \sum s_i^2}{s_t^2} \right] \quad (j)$$

where s_i^2 is the variance of scores on item i and s_t^2 is the variance of the total test scores. Although the Kuder-Richardson formulas are applicable only when test items are scored "0" (wrong) or "1" (right), Cronbach's alpha is a general formula for estimating the reliability of a test consisting of items on which two or more scoring weights are assigned to answers.

Although Cronbach's alpha has the advantage of identifying which items are or are not contributing to the overall reliability, the disadvantages is that each and every item has to be individually assessed for variability.

8.2.5 Analysis of Variance in Estimating Reliability

The well-known analysis of variance techniques can be applied to assess the internal consistency test reliability. This technique can be employed for establishing r_{tt} one for the situation in which the items are dichotomously scored (right-wrong, agree-disagree), and also one in which the items are scored on an interval scale. We illustrate the method only with the dichotomous data.

The method presented here is due to Cyril Hoyt, a great statistician and is popularly known as **Cyril Hoyt method**. We illustrate this method by an example.

Example 8.6: A specialist has created a four-item test of shyness and given it to a random sample of five subjects. The test is scored only on the basis of whether the item indicates shyness or whether it does not. It was scored as 1 for shyness and 0 for no shyness (see data below). Using ANOVA technique, estimate the internal consistency reliability of the test.

Subjects	Items				Row sum
	a	b	c	d	
1	1	1	1	1	4
2	1	1	1	1	4
3	1	1	1	0	3
4	1	1	0	0	2
5	1	0	0	0	1
Column sum	5	4	3	2	14

The following steps are involved in the computation of reliability coefficient.

1. Compute the correction factor C for n =number of scores=20

$$C = \frac{(\sum x)^2}{n} = \frac{14^2}{20} = 9.80$$

2. Find the total sum of squares SS_t by adding the squares of the score values and subtracting C:

$$SS_t = (1^2 + 1^2 + \dots + 0^2) - C = 14 - 9.8 = 4.2$$

3. Get the sum of squares between columns (SS_{items})

$$SS_{\text{items}} = \frac{5^2}{5} + \frac{4^2}{5} + \frac{3^2}{5} + \frac{2^2}{5} - 9.8 = 10.8 - 9.8 = 1.0$$

4. Find the sum squares between rows (SS_{subjects}):

$$SS_{\text{subjects}} = \frac{4^2}{4} + \frac{4^2}{4} + \frac{3^2}{4} + \frac{2^2}{4} + \frac{1^2}{4} - 9.8 = 11.5 - 9.8 = 1.7$$

5. Get the $r \times c$ (residual) by subtraction:

$$SS_{r \times c} = SS_t - SS_{\text{items}} - SS_{\text{subjects}} = 4.2 - 1.0 - 1.7 = 1.5$$

6. Now construct the ANOVA table

Sources	SS	df	MS	F
Between columns (items)	1.0	3	0.333	–
Between rows (subjects)	1.7	4	0.425	3.4
Residual ($r \times c$)	1.5	12	0.125	
Total	4.2	19	–	–

$$\text{Note: } F = \frac{MS_{\text{subject}}}{MS_{\text{residual}}}$$

The tabulated value of F at 5% level with 4 and 12 df is 3.26. Hence our calculated F is significant. We can proceed to calculate the reliability.

$$r_{tt} = 1 - \frac{MS_{\text{residual}}}{MS_{\text{subject}}} = 1 - \frac{.125}{.425} = 0.706$$

This test is showing fair to low reliability.

8.2.6 How to Increase Reliability

The estimated reliability of a test can be increased by the following techniques:

- Increase the length of a test by adding more items. The longer the test is, the higher the likely reliability. One should only add items that have been selected from the same population of items.
- You can achieve enhanced reliability through improved investigator consistency by using only well-trained supervised and motivated persons to conduct the research.
- Increase the variance by making the test more discriminating. Add items that separate persons along a continuum, with some items that only the top scores will get right, some only the worst scores will get wrong, and so on up and down the line. Get rid of items that everybody gets right (as well as those that everyone gets wrong), since they are not providing any power to discriminate. A 2-item test obviously cannot yield as much variation as a 100-item test, nor will the 2-item test provide much information on the discriminating power of the items, other than at best perhaps sorting the above average from the below average.
- Increase the size of the sample being tested. No matter how many items you have or how well your items discriminate, if there are not enough subjects of varying ranges of ability, the reliability of the test can be severely underestimated.

8.3 Standard Error of Measurement

The reliability of a test may be expressed in terms of the **standard error of measurement** (s_e), also called the **standard error of a score**. This score is particularly well suited to the interpretation of individual scores. For many testing purposes, it is therefore more useful than the reliability coefficient. The standard error of measurement can be easily computed from the reliability coefficient of the test by simple rearrangement of the Rulon formula (g):

$$s_e = s_t \sqrt{1 - r_{tt}} \quad (k)$$

in which s_t is the standard deviation of the test scores and r_{tt} the reliability coefficient, both computed on the same group.

The standard error of measurement is a useful statistic because it enables an experimenter to estimate limits within which the true scores of a certain percentage of individuals (subjects) having a given observed score can be expected to fall, assuming that errors of measurement are normally distributed. According to the normal law of probability, about 68% (more precisely 68.27%) of a group of individuals will have true scores falling within ± 1 standard error of the measurement of the observed score. Likewise, about 95% of a group of individuals having the same observed scores will have true scores falling within ± 2 standard error of the measurement of the observed score. And virtually all (99.72%) will have true scores falling within ± 3 standard error of the measurement of that observed score.

As an example, assume that the standard deviation of the observed scores on a test is 10 and the reliability coefficient is 0.90; then

$$s_e = 10\sqrt{1 - .90} = 3.16$$

So if a person's observed score is 50, it can be concluded with 68.27% confidence that this person's true score lies in the interval $50 \pm 1(3.16)$, i.e. between 46.84 and 53.16. If we want to be more certain of our prediction, we can choose higher odds than the above. Thus we are virtually certain (99.72%) that this individual will have his true score in the range 40.52 and 59.48 resulting from the interval $50 \pm 3(3.16)$.

As formula (k) shows, the standard error of measurement increases as reliability decreases. When $r_{tt} = 1.0$, there is no error at all in estimating an individual's true score from the observed score. When $r_{tt} = .00$, the error of measurement is a maximum and equal to the standard deviation of

observed score. Of course, a test with a reliability coefficient close to 0.00 is useless because the correctness of any decisions made on the basis of the scores will be no better than chance.

How do we judge the usefulness of a test in terms of reliability coefficient? The answer to this question depends on what one plans to do with the test score. If a test is used to determine whether the mean scores of two groups of people are significantly different, then a reliability coefficient as low as 0.65 may be satisfactory. If the test is used to compare one individual with another, a coefficient of at least 0.85 is needed.

8.4 Validity and its Measurement

For the test to have any practical use, it must not only be reliable, but it also must be valid. A traditional definition of validity of a measuring device is the extent to which it measures what it was designed to measure. That is, if a test has been designed to measure musical aptitude, for example, a valid test measures just that, and not some other extraneous variable(s). When you step on the scales, you want to know your weight, not your IQ or some other unknown quality. In scientific usage, a measurement of a given phenomenon (as designated by a given concept) is viewed as a valid measure if it successfully measures the phenomenon (Phillips, 1971: 197, *cf.* Baily, 1982: 66). It seems clear that the definition of validity has two parts: (1) that the measuring device is actually measuring the concept in question and not some other concepts; (2) that the concept is being measured accurately. Obviously, one can have the first without the second but not vice-versa (the concept cannot be measured accurately if some other concept is being measured).

To clarify the concept more clearly, assume that we wish to measure IQ, and that Afra certainly has some actual level of IQ (say 110). We can construct a test that measures her IQ, but measures it with some degree of error. That is it measures her intelligence as 100 when it is really 110. No doubt, such a test is measuring the right concept (IQ) and has some potential for being a valid measurement if we are able to refine it so as to remove the error in measurement. On the other hand, if we wish to measure Afra's intelligence, we may devise a test which does not actually measure intelligence at all. Such a test can never be a valid measure of intelligence as it is measuring some other concept (such as her skill, or may be measuring nothing at all), rather than measuring intelligence as we had planned. The difficulty is that this test may still yield a numerical score, and could conceivably even yield a score of 110 for Afra. However, it still is not valid because it is not measuring the desired phenomenon.

Thus, it obviously becomes very important to be able to assess the validity of tests, as all of them will yield some sort of score, but this does not necessarily mean that the measurement is valid. By definition, if a measure is valid, it will be accurate every time, and thus be reliable also, but the converse is not true. That is a reliable measure may not be valid. Another way of putting the same statement is that reliability is a necessary condition but not a sufficient condition for validity.

Validity of measurement can be viewed broadly in two ways: internal validity and external validity. **Internal validity** seeks to answer whether a difference exists at all in any given comparison. It asks whether or not an apparent difference can be attributed to same measurement artifact. If one does not encounter contradictions in the data within a given experiment, then the study is said to have internal validity. Contradictory findings signal the absence of internal validity. **External validity** is the problem of interpreting the difference, the problem of generalization of the experimental research. Even if internal validity is present, the findings are not said to have external validity unless they are held to be valid for additional (external) situations besides the original study that generated the findings.

Many forms of validity are mentioned in the research literature, and the number grows as we expand the concern for more scientific measurement. One accepted classification consists of four major forms: (a) face validity (b) content validity, (c) criterion-related validity, and (d) construct validity. We discuss these in brief.

8.4.1 Face Validity

The most basic method of testing for validity is to carefully examine the measure of a concept in the light of its meaning and to ask oneself seriously whether the measuring device really seems to be measuring the underlying concept. This form of careful consideration and examination is referred to as **face validity**. Face validity is based on the general appearance of the test, and as such a test will have more face validity if its look and feel are consistent with the construct under study. From a testing perspective, it is really important to have a test that looks on the surface like it can tap the characteristic under study. This surface appearance of the test may motivate the test takers, especially when the test produces an assumption of relevance. For example, a test for measuring a person's aptitude for banking might be more realistic and motivating if it asks questions about balance sheets and interest payments, rather than, say, the batting average of a certain cricket player. What is important in validity

context is that a test has face validity if the items are reasonably related to the perceived purpose of the test.

Face validity apparently reflects the content in question. It is simply assessed by asking other people whether the measure seems to be getting at the concept that is the focus of attention. In other words, people, possibly those with experience or expertise in a field, might be asked to act as judges to determine whether on the face of it the measure seems to reflect the concept concerned. Face validity thus is essentially an intuitive process that judges whether the measuring instrument arrives at the concept adequately.

8.4.2 Content Validity

Content validity seeks to answer whether the empirical indicators (tests, scales, questions, etc) fully represent the domain of the underlying concept being studied. In this sense, content validity concerns the degree to which the content of a set of items adequately represents the universe or domain of all relevant items under study. To the degree that the items reflect the full domain of content, they can be said to be content-valid.

Determination of content validity is judgmental and can be approached in several ways. Two of the approaches are:

- The designer may determine it through a careful definition of the topic of concern, the items to be scaled, and the scales to be used. This logical process is often intuitive and unique to each research designer.
- One may use a panel of persons to judge how well the instrument meets the standards.

The major problems with content validity arise when

- There is no consensus about the definition of the concepts to be measured
- The concept is multi-dimensional one consisting of several sub-concepts
- The measure is lengthy and complex.

8.4.3 Criterion-related Validity

Criterion validity, variously called pragmatic validity, concurrent validity or predictive validity, involves multiple measurement of the same concept. The term **concurrent validity** has been used to describe a measure that is valid for measuring a particular phenomenon at the same time that the

measurement is administered, while **predictive validity** relates to future performance on the criterion. These two measures thus differ only in a time perspective. Predictive validity is concerned with how well a scale or instrument can forecast a future criterion and concurrent validity with how well it can describe a present one. Look at the following cases:

- An opinion questionnaire is designed to forecast the outcomes of a bank employees' union election. Our concern is to assess how correctly the questionnaire forecasts the results of the election. The more the questionnaire accurately forecasts the results, the more predictive validity the questionnaire possesses.
- An observational study is conducted to classify families as having low, medium and high income. The degree, to which the study can correctly classify these families, represents the concurrent validity.
- A questionnaire is designed to identify psychiatrically disturbed persons. The validity of the questionnaire might be determined by comparing its diagnoses with those made by a psychiatrist on the basis of clinical investigation. The two approaches respectively lead to predictive validity and concurrent validity.

The opinion, income, and psychiatric condition in the above examples are the criteria which we employ to measure the validity of the measuring instruments.

Ratings of performance, units produced in a certain period of time, amount of sales and the number of errors made in a typing document are a few more examples of criteria.

Once the specification of the test is made and criterion scores are obtained, they may be compared in some way. The usual approach is to correlate them using product-moment correlation. For example, we might correlate test scores of a group of college students and the grade point average that these students made during their first year of college and regard this coefficient as **validity coefficient**. The validity coefficients usually tend to be much lower than the reliability coefficient (in the range .40-.60 with a median value of .50).

8.4.4 Construct Validity

Criterion-related validity is based upon getting some empirical evidence (such as scores, income, ratings) to serve as the basis for judging that what is being measured really measures what it is supposed to measure (predictive capacity). This is an empirically based form of validity in

which some observable evidence can be used to confirm the validity of a measure. However, when there is neither a criterion nor an accepted universe of content that defines the quality being measured, then criterion-related validity cannot be used to test for validity. The difficulty in applying rigorous and objective tests of validity in attitude measurement, for example, arises from the fact that such measurement is invariably indirect in the sense that the attitude is inferred from the verbal statements or responses. An attitude is an abstraction and in consequence it is generally impossible to assess its validity directly. An approach for this situation is that of **construct validity**. Attitude test, aptitude and personality tests generally concern concepts that fall in this category.

Construct validity is based on forming hypotheses about the concepts that are being measured and then on testing these hypotheses and correlating the results with the initial measure. Here is a 3-step procedure to test the construct validity of measures:

- Specify first the theoretical relationship between the concepts themselves.
- Examine the empirical relationship between the measures of the concepts.
- Interpret the empirical evidence in terms of how it clarifies the construct validity of the particular measure.

Suppose that we develop two indices of measuring social class and label these indices as index-1 and index-2. Assume that we have a theory that contains a hypothesis which states that an inverse relationship exists between social class and crime—as social class increases, crime decreases. Assume that this hypothesis has been tested by measuring social class by index-1, and has been established by statistical tests. Construct validity consists in replacing index-1 by index-2 in the theory and retesting the entire theory. If we get the same results for the whole theory (especially for the hypothesis containing index-2) as when we used index 1 to measure social class, then we say that the new measure (index-2) has construct validity.

The types of validations discussed so far can be seen as an accumulation, with each of the subsequent types of validation including all of the elements of all former types, along with some new features. In other words, construct validation requires more information than criterion validation, and criterion validation requires more information than content or face validation. For this reason, construct validation is often said to be the strongest kind of validation procedure. In summarizing the various

types of validity testing for their effectiveness in social research, Zeller and Carmines conclude that construct validity is the most useful and applicable to the social sciences.

To conclude, it must be noted that the reliability and validity of measurement are not invariant characteristics. They are always specific to a particular population, time and purpose. In any given situation, the researcher therefore has to decide what degree of unreliability and invalidity he will consider as acceptable.

8.5 Constructing Measurement Scales

Scales are central to the subject of measurement in social research. A scale refers to an item or set of items (indicants) for measuring some characteristic or property, such as an attitude. Scaling is a procedure for the assignment of numbers to a property of objects to impart some of the characteristics of numbers to the properties in question. Thus, you may devise a scale to measure the durability (property) of paint. In another context, you may judge a person's supervisory capacity (property) by asking a peer group to rate that person on various questions (indicants) that you create. You may also measure the IQ score (property) of a person asking him pre-designed questions (indicants).

A scale generally represents a single complex concept, or construct, that combines multiple indicators into a common composite measure. The reason why scales are widely used in the social sciences is that many of the concepts which social researchers want to study cannot be measured with a single indicator. Concepts like religiosity are too multidimensional to be measured with a single item. When such concepts are of interest, a scale or an index can be developed to try to incorporate the many facets of the abstract variable into a set of indicators representing its operational definition.

Scales are of two types: rating or ranking. **Ratings scales** have multiple response categories and are also known as **category scales**. They are used when respondents score an object or attitude without making a direct comparison to another object or attitude. For example: the respondents may be asked to rate the styling of a new shoe or quality of a garment on a five-point scale: excellent, good, average, below average, and poor or frequency of visits of a community health worker: always, often, occasionally, rarely and never. **Ranking scales** make comparisons among two or more attitude or objects. The respondents under this method compare two or more objects and make choices among them. For example, respondents can express his or her attitude by making a choice between

two brands of car. In such scaling, one may also follow the rank ordering. Under this approach, the respondents are asked to rank their choices according to the importance of the objects.

Scales can again be unidimensional or multidimensional. With a **unidimensional scaling**, one attempts to measure one attribute of the respondent or object. One measure of employee potential is promotability. The manager may decide to identify the employee's technical performance for promotion. It is a single dimension. Several items may be employed to measure this dimension, and by combining them into a single measure, a manager may place employees along a linear continuum of promotability. **Multidimensional scaling** recognizes that an object may be better described in an attribute of n dimensions rather than on a unidimensional continuum. The employee promotability variable might be better expressed by three distinct dimensions—managerial performance, technical performance, and team-work.

We discuss below a few attitude rating scales.

8.5.1 Simple Category Scale

In its most basic form, attitude scaling requires that an individual agrees or disagrees with a statement or responds to a single question. For example, respondents on a political issue may be asked whether they agree or disagree with the following statements:

Statements	Agree	Disagree
1. Election Commission should be reformed for free and fair election.	1	0
2. A Government employee will not be eligible for participating in national election before he/she retires from his/her job.	1	0
3. Chief Election Commissioner must be chosen from among the retired chief justices.	1	0

Each of the respondents selects either a positive response (agree) or a negative response (disagree) which most accurately describes his/her attitude towards the opinion. A person's score is computed by summing up the number of questions he or she answers in certain ways. In the present context where the responses have been coded 1 and 0, a person who agrees with all the opinions makes a maximum score (3 here) on the scale and a person who disagrees with all opinions receives a score 0.

Because this type of self-rating scale merely classifies respondents into one of two categories, it has only the properties of a nominal scale.

Single Response Scale

Many people believe that attitudes vary along a continuum. Thus the purpose of an attitude scale is to find an individual's position on a continuum of multiple choices. When there are multiple options for the rater and only one response is sought, this scale can be employed. These options provide the respondents more flexibility in the rating operation. We can ensure more information if the categories are ordered according to a descriptive or evaluative dimension. An example of this scale is as follows:

- How frequently the Family Welfare Assistant visits you?

The responses to these questions are provided below:

1. Never
2. Rarely
3. Sometimes
4. Often
5. Very often

Each of the respondents selects one response category which most accurately describes his or her opinion about the statement reflecting the concept (frequency of visits) being measured. The data obtained by this rating scale is ordinal in nature.

The above category scale is more sensitive and more informative than a scale with only two response categories. It is however to be noted that wording is an important factor in the usefulness of these scales.

Consider another example.

- Which of the following newspapers do you read **most** often for financial news?
1. The New Nation
 2. The Observer
 3. The Independent
 4. The Times
 5. The Daily Star

Multiple Response Scale

This scale allows the rater to select one or several alternatives from among the multiple options. Here is an example:

- Check one or more of the following sources you consulted while designing your new apartment complex:
 1. Online planning services
 2. Magazines
 3. Independent builders
 4. Developer's model/plans
 5. Designer
 6. Architect
 7. Others (specify).....

Here too, a respondent's total score is the summation of the scores he obtains in an interview of expressing his/her opinion.

8.5.2 Summated Scale

A summated scale involves a number of statements with several response categories indicating the intensity of agreement or disagreement. These statements represent either positive or negative, but not neutral or equivocal attitudes of the respondents about the issue involved. Each response category is given a numerical score indicating the degree of agreement. All the scores of the statements are then matched in their positive or negative direction, and the scores are summed. The added score values represent the attitude of a respondent on an issue and the respondent's position with reference to other respondents. This type of scale frequently used in the measurement of social attitude was first devised by Rensis Likert. Hence it is also known as **Likert scale**.

The essence of Likert's technique is to increase the variation in the possible scores by coding from "strongly agree" to "strongly disagree" instead of merely "agree or disagree". A scale score is computed for each respondent. If person "A" makes a very high score and person "B" a very low score, their answers to all questions can be compared to see if there is any single question that the persons with the two very different scores answered identically. Such a question obviously does not discriminate well between discrepant scores on the concept, and so would be eliminated from the scale. For example, if person "A" had a score of 60 and person "B" a score of 5, but both had identical answer on question 7, then question 7 would be dropped from the scale.

We illustrate this scale by an example below:

- The Internet is superior to traditional libraries for comprehensive search

Each respondent was asked to react on the above issue and his or her response on this particular issue was recorded in one of the five-levels of agreement. These levels of agreement along with pre-assigned scores are shown below:

Strongly agree	Agree	No opinion	Disagree	Strongly disagree
(5)	(4)	(3)	(2)	(1)

The numbers indicate the value to be assigned to each possible answer with 1, the least favorable impression of Internet superiority and 5, the most. These values are normally not printed on the instrument. We do it here just to indicate the scoring system.

A Likert scale may include several scale items to form an index. We cite another example with five statements below that describe how marketing executives of private sector enterprises feel about designing a sales-force of an enterprise.

	Strongly agree	Agree	No opinion	Disagree	Strongly disagree
Steps	(5)	(4)	(3)	(2)	(1)
1. Determining sales-force objectives.	5	4	3	2	1
2. Determining sales-force strategy.	5	4	3	2	1
3. Determining sales-force structure.	5	4	3	2	1
4. Determining sales-force size.	5	4	3	2	1
5. Determining sales-force compensation.	5	4	3	2	1

In an abortion issue, one can similarly think of several statements that best describe the issue. These may, among others, be

- **Statement 1:** Abortion should be legalized if women agree.
- **Statement 2:** An unborn child has equal human right as anyone else in the society.
- **Statement 3:** If a woman does not want any more children, she has a natural right to an abortion.
- **Statement 4:** All women should use contraceptives in order to avoid any unwanted birth that she wants to abort.

Each statement is assumed to represent an aspect of a common attitudinal domain or continuum. Ordinarily, between 20 and 25 properly constructed

statements about an attitude object would be required for a reliable Likert scale. The total score is the summation of the weights assigned to an individual's response, which represent the attitude of a respondent on an issue.

Suppose there are 25 statements. Then the minimum score of an individual will be $25 \times 1 = 25$ indicating a most unfavorable attitude toward the issue involved and the maximum score will be $25 \times 5 = 125$, indicating the most favorable attitude, the intermediate or neutral score being $25 \times 3 = 75$.

One important point is in order. Before assigning the score or weights to possible answers, it is necessary to take the direction of the items into accounts. This means that if an item is worded positively with assigning a score '5', for 'strongly agree', then for another item which is worded negatively the score '5' should be assigned for 'strongly disagree'. It is not important whether a '5' is assigned to 'strongly agree' or 'strongly disagree', what is important is to score the responses consistently in terms of the attitudinal direction they indicate.

The basic procedure for Likert scaling is as follows:

- (a) Write an adequate number of statements (or questions) thought to measure the attitude to be scaled;
- (b) Select a sample of respondents, representative of the population on which the scale will be used;
- (c) Ask each respondent to read each statement and to state the level of his agreement with it, using a five-point scale;
- (d) Code all responses so that a higher score on a particular item indicates a stronger agreement with the attitude being scaled (code 5 for either strong agreement with a positive statement or strong disagreement with a negative one, and code 1 for strong disagreement with a positive statement or strong agreement with a negative one);
- (e) Compute a scale score for each respondent by summing his or her scores on all questions;
- (f) Analyze responses and select for the scale the items most clearly differentiating between the highest and the lowest scores;
- (g) Re-compute final total score.

In analyzing and re-computing the final score, the researcher can array these total scores and select some portion representing the highest and the lowest total score, say the top 25% and the bottom 25% of the scale scores.

Now compute the median scores on each item for the highest 25% and the lowest 25% of scale scores. If any question has the same median score for both the highest and the lowest groups, it can be eliminated from the scale. Only questions that reveal widely different median scores for the highest and the lowest groups will be retained.

The Likert scale has many advantages that account for its popularity. It is easy and quick to construct. Each item that is included has met an empirical test for discriminating ability. Since respondents answer each item, it is probably more reliable and it provides a greater volume of data than many other scales. The Likert's scales also help us compare one person's score with a distribution of scores from a well-defined group. They are also useful when we expect to conduct an experiment or undertake a program of change or improvement. We can measure attitudes before and after the experiment or judge whether our efforts have had the desired effects. This scale produces interval data.

8.5.3 Thurstone Scale

Thurstone scale is associated with the name of L. L. Thurstone. In this scale a number of statements, whose position on the scale has been determined by judges, is used. The position is determined by the method of equal-appearing intervals. The procedure is as follows:

At first, a reasonably large number of statements relating to the attitudes to be studied are collected. A large number of judges work independently to classify these statements into 11 groups. In the first group, all the statements which are most unfavorable to the issue are placed. The next unfavorable statements are placed in the second group, and so on. The statements of the 11th group are considered most favorable. The sixth position (the middle one) on this continuum is the point at which the attitude is neutral. The first group is given a score 11, and the 11th group the score 1.

The scale-value of a statement is computed as the mean or median position to which it is assigned by the judges. Statements which are ambiguous, vague or irrelevant and over which judges differ widely are discarded. Finally, the scale is prepared by taking into account the evaluated statements that spread out evenly from one extreme to the other.

Suppose that a statement (say *A*) has been distributed to 50 judges into 11 categories as follows:

Statement (x)	Number of judges (f)	Product fx
1	1	1
2	6	12
3	9	27
4	20	80
5	9	45
6	5	30
7	4	28
8	3	24
9	2	18
10	1	10
11	0	0
Total	50	275

We treat the statements as variable values (x) and the number of judges as frequencies (f), and calculate the mean value as follows:

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{275}{50} = 5.5$$

The median score is computed to be 5. If we choose median, then the scale value of the statement A is 5. In a similar manner, every item is assigned a numerical score.

Thurstone scale consists, thus, of a series of statements whose positions have been determined neutrally by the judges. At the time of administration of the scale questionnaire, the respondents are asked to check the statements with which they agree. Like Likert scale, the scale values are not shown in the questionnaire and the statements are arranged randomly.

8.5.4 Graphic Rating Scale

A graphic rating scale presents respondents with a graphic continuum. The respondent checks his or her response at any point along a continuum. Typically a respondent's score is determined by measuring the length (in millimeters) from either end point of the graphic continuum. The results are usually interval data. Other graphic rating scales use picture, icons, or other visuals to communicate with the raters. Here is an illustration of a graphic scale:

- How often should a manager supervise his business?

The responses to this question may be put on a graphic scale as under

×

Never -----Always

Here × is the position along the line that best describes your judgment

8.5.5 Cumulative Scale

In cumulative scale, a respondent is given a number of questions to express his or her agreement or disagreement on an issue. The items are arranged in such a way that a respondent who responds favorably to item 2 also replies favorably to item 1 and one who replies favorably to item 3 also replies favorably to items 1 and 2, and so on. In other words, it assumes a cumulative set of scores. Therefore, the individuals who answer favorably have higher total score than those who answer unfavorably. The score of an individual is computed by counting the number of items he answers favorably. His scores indicate for him a particular position on the scale. The intervals between the positions may not be equal. The items may be arranged from favorableness to unfavorableness in a systematic manner or may be randomly selected.

The cumulative type of scale was successfully used by Bogardus first for which it is also known as **Bogardus's social distance scale**. The main purpose of this scale is to measure the attitude towards a particular ethnic group or groups. A number of suggested relationships may be listed, to which members of an ethnic group (for example) may be accepted. The respondent is to indicate as to which racial group is to be accepted by him for each of the specified relationships. The attitude is measured by the closeness of relationship that a respondent is willing to accept or the social distance that he likes to maintain. A Bogardus-type scale is illustrated by an example below:

Relationship I would accept	Muslim	Hindu	Buddhist	Christian
1 Marriage	1	1	1	1
2 Friendship	2	2	2	2
3 Neighbor	3	3	3	3
4 Job in same occupation	4	4	4	4
5 Partnership in business	5	5	5	5
6 Visitor in my country	6	6	6	6
7 Expulsion from my locality	7	7	7	7

The respondent is to circle each of the seven categories to which he is willing to accept a particular group. The respondent's first reactions can be known by this. For a group, if a respondent circles '3', he is very likely to circle '4' and '5' for the same group. If the respondent does not circle '3',

he will probably not circle '1' and '2', for these indicate even closer relationship for the same group. The seven categories indicate gradually increasing social distance. The score for the respondent will be the number of items he has circled. From this score, it is possible to ascertain which of the items the respondent has chosen.

In the Bogardus-type scale, the respondent has to indicate his first feeling. He has to give his reaction to each race as a group, not an individual member of the group. The scale distance can also be calculated mathematically. In order to do this, weights are attached to different categories relationships. Thus, if there are only 5 categories, the weights such as 1, 2, 3, 4, and 5 can be assigned to the first 5 categories respectively. The following procedure is generally adopted for the measurement of social distance:

- Place the weights and percentage response for each category in rows
- Multiply the percentage response by its weight
- Add up the product, and this will be the required social distance.

One problem that have been raised with this form of scale is that there is no way to determine the actual distance between the various points on the scale, and some points seem to be at a greater distance from the point next to them than others further away. The scores, however, are treated as equidistant. Nevertheless, such a scale – and there are many versions – may be useful if we are studying attitudes toward groups of “others”.

8.5.6 Guttman Scale

In statistical surveys conducted by means of structured interviews or questionnaires, a subset of the survey items having binary (e.g., YES or NO) answers forms a **Guttman scale**, (named after Louis Guttman) if they can be ranked in some order so that, for a rational respondent, the response pattern can be captured by a single index on that ordered scale. In other words, on a Guttman scale, items are arranged in an order so that an individual who agrees with a particular item also agrees with items of lower rank-order. For example, a series of items could be

1. I am willing to be near ice cream;
2. I am willing to smell ice cream;
3. I am willing to eat ice cream; and
4. I love to eat ice cream.

Agreement with any one item implies agreement with the lower-order items. This contrasts with topics studied using a Likert scale or a Thurstone scale. The method of scaling devised by Guttman is also called **scalogram analysis**. A well known example of a Guttman scale is the **regardus Social Distance Scale**, which is as follows:

1. Are you willing to permit immigrants to live in your country?
2. Are you willing to permit immigrants to live in your community?
3. Are you willing to permit immigrants to live in your neighborhood?
4. Are you willing to permit immigrants to live next door to you?
5. Would you permit your child to marry an immigrant?

The concept of Guttman scale likewise applies to series of items in other kinds of tests, such as achievement tests, that have binary outcomes. For example, a test of math achievement might order questions based on their difficulty and instruct the examinee to begin in the middle. The assumption is if the examinee can successfully answer items of that difficulty (e.g., summing two 3-digit numbers), s/he would be able to answer the earlier questions (e.g., summing two 2-digit numbers). Some achievement tests are organized in a Guttman scale to reduce the duration of the test.

By designing surveys and tests such that they contain Guttman scales, researchers can simplify the analysis of the outcome of surveys, and increase the robustness. Guttman scales also make it possible to detect and discard randomized answer patterns, as may be given by uncooperative respondents.

A hypothetical, perfect Guttman scale consists of a unidimensional set of items that are ranked in order of difficulty from least extreme to most extreme position. For example, a person scoring a "7" on a ten item Guttman scale, will agree with items 1-7 and disagree with items 8, 9, 10. An important property of Guttman's model is that a person's entire set of responses to all items can be predicted from their cumulative score because the model is deterministic.

An important objective in Guttman scaling is to maximize the reproducibility of response patterns from a single score. A good Guttman scale should have a coefficient of reproducibility (the percentage of original responses that could be reproduced by knowing the scale scores used to summarize them) above .85. Another commonly used metric for assessing the quality of a Guttman scale, is Menzel's coefficient of scalability and the coefficient of homogeneity (Loevinger, 1948; Cliff,

1977; Krus and Blackman, 1988). To maximize unidimensionality, misfitting items are re-written or discarded.

A scale is said to be **unidimensional** if the responses fall into a perfect **pattern** in which endorsement of the item reflecting the extreme position results also in endorsing all items that are less extreme. With the Guttman technique, the 'perfect' implies that a person who replies a given question favorably will have a higher score than a person who answers it unfavorably. In this situation the number of items endorsed by a respondent giving a complete picture as to which items he agreed and disagreed with, serves as his score. If some combination of scores other than the desired combination forms a particular scale score, it is considered to be in error. If there are many scaling errors, or exceptions to the desired pattern, the scale is inadequate.

The attainment of higher degree of unidimensionality is the major concern of the Guttman scale. However, the Guttman scale belongs to the broad category of cumulating scaling. It is cumulative in the sense that the combination responses required to make a particular score includes the responses to all questions required to make the next lower score, plus the response to one additional question, in a stepwise fashion.

According to Guttman, a 'universe of content' can be considered to be unidimensional only if it yields a perfect or nearly perfect cumulative scale. Suppose a survey is conducted among 100 respondents on the opinion of a new brand of shirt labeled 'union'. We may have developed a preference scale of 4 items as follows:

Items	Opinions
1.	'Union' is fairly attractive
2.	The overall appearance of 'union' is acceptable to me.
3.	'Union' is more preferable to me than other styles in the market
4.	I am convinced to buy 'union' next time

A score enables one to determine as to which item is endorsed by the respondent. Person with a score of 3, for example, should disagree with item 4 but agree with all others. A score of 4 indicates all statements are agreed to and represents the most favorable attitude. According to the scalogram theory, this pattern confirms that the universe of content (attitude toward an issue under study) is scalable. How is a score assigned to an individual? A commonly used general solution is to assign to each individual a score equal to the number of items he endorses.

The respondents who participated in the survey indicate whether they agree or disagree with the opinions. If these items form a unidimensional scale, the response pattern will form a perfect cumulative scale of the following type:

Scalogram Response Pattern (ideal scale structure)

Item					
4	3	2	1	Respondent score	No. of respondents
×	×	×	×	4	20
–	×	×	×	3	22
–	–	×	×	2	6
–	–	–	×	1	10
–	–	–	–	0	7
Total	.	.	.	–	65

× = Agree, – = Disagree

In the case of non-perfect pattern, the respondents will act differently. Such a pattern will be designated as non-scale pattern. The number of respondents doing so will be remainder of those who followed scale pattern as in the above table. If there are 100 persons participating in the study and 65 responded according to the scale pattern, then the remaining 35 did respond differently resulting in a non-scale pattern, which we categorize as errors. The responses of this pattern may be as follows:

Scalogram Response Pattern (Non-scale Structure)

Item				
4	3	2	1	No. of respondents who are in error
–	–	×	–	10
–	×	–	×	9
×	–	–	×	9
×	×	–	×	7
Total				35

× = Agree, – = Disagree

In the Guttman technique, the perfect scale implies that a person, who answers a given question favorably, will have a higher total score than a person who answers it unfavorably. In the first table, the four response patterns form a scalar pattern. If all the responses provided by the respondents form a scalar pattern (in which case there will be none to form the second table), the items are said to form a perfect scale and the resulting scale has been called by Guttman a "perfectly reproducible" one. In the second table, the four rows constitute the non scale patterns and each of these patterns is associated with error.

The Guttman scale is analytically complex, apart from the fact that there is no guarantee that the various items will scale, and even if they do so, the universe of content may be narrow in coverage. This method is more appropriate for scaling ordered behavior than less structured and broad-based attitudes (Ghosh, 2003:173).

The reproducibility of Guttman score can be examined by what is known as **coefficient of reproducibility (C_R)**. The higher the value of the coefficient, the higher the proportion of scores we can reproduce accurately. The coefficient is computed by subtracting the proportion of responses that are errors from 100 per cent (all responses). The formula to be used is

$$C_R = 1 - \frac{\text{Number of respondents in error (E)}}{\text{Number of respondents (N)} \times \text{Number of items (n)}} = 1 - \frac{E}{N \times n}$$

Another way of expressing C_R is to say that it is the proportion of responses that can be correctly predicted from the individuals' total score. A coefficient of reproducibility can also be calculated for each item individually as the proportion of responses on that item that are correctly predicted; C_R is the simple average of these item coefficients.

In the foregoing example, $E=35$, $N=100$, $n=4$, so that

$$C_R = 1 - \frac{35}{400} = 91.25\%$$

Any coefficient of reproducibility over 0.90 is supposedly adequate to indicate scalability and the ability to reproduce responses to the various items from knowledge of the total score. However, as Edwards (1957: 191) notes, a C_R of 0.90 is not a sufficient condition for the scalability of a set of statements.

1.5.7 Semantic Differential

The semantic differential technique was developed by Osgood, Suci and Tannenbaum (1957). Its main objective is to examine the meaning of certain concepts, e.g., church, wife, socialism, and so on. In a sense, it is an attitude scale. The groups of subjects may be asked to rate a given concept on a series of bipolar rating scales. Generally, a seven-point scale is utilized, whose two ends will use adjectives which are opposite (called bipolar terms), e.g., good-bad, complex-simple, passive-active, friendly-unfriendly and so on. The response categories consist of seven categories ranging from one extreme to the other with the middle category representing neutral. The seven positions for each item are assigned score

from 1 through 7. Each respondent is then asked to indicate his or her perception about the object by placing a check mark, that is, by choosing a score on each item that describes the object with respect to the bipolar terms for that item. The individual scores for the items are then summed, and his total score is the measure of the respondent's perception about the object.

The semantic differential is similar to the Likert scaling categories ranging from strongly disagree to strongly agree, except that in the semantic differential only the two end categories have names. The middle categories simply have a blank space, or sometimes a number. Also the two end categories are not strongly agree—strongly disagree, but rather a pair of opposite adjectives (bipolar terms) thought to express the subject's feelings about the concept. For example, opinion on the issue of abortion or early marriage may be expressed as good or bad as extremes. The format for response categories may be displayed as below:

ABORTION
Good—————Bad

Osgood has investigated the correlations between the scores given to a set concept on different bipolar scales by conducting a series of factor analysis. The three main factors are individual's **evaluation** (e.g., social-unsocial, successful-unsuccessful), **potency** (e.g., weak-strong, heavy-light) and the individual's **activity** (e.g., active-passive, slow-fast) of the object or concept. In other words, people tend to react to various concepts largely (but not entirely) in terms of these dimensions.

The semantic differential makes the measurement and comparison of various objects or concepts possible. In order to form an attitude scale, what we need is to decide the description of the issue to be studied, and to choose suitable objective pairs for it. A respondent's total score is the measure of his attitude.

8.5.8 Stapel Scale

The Stapel scale is used as an alternative to semantic differential, especially when it is difficult to find bipolar adjectives that match the investigative question. The scale places a single adjective in the center of an even number of numerical values (for example, ranging from +3 to -3). It measures how close or how distant from the adjective a given stimuli is perceived to be. Table below illustrates a Stapel scale item used in a measurement of attitude of the student toward a class teacher:

importance of the variable in measuring each factor. The factor analysis program derives the factors so that they are uncorrelated with each other (i.e. independent of each other). The factor loadings can be presented in the following manner:

Variables (or items)	Factor-1 (Scale-1)	Factor-2 (Scale-2)
Height	0.58	0.48
Weight	0.68	0.67
Occupation	0.54	0.56
Education	0.47	0.77
Income	0.67	0.57

Note: The factor loadings are hypothetically assumed

It is up to the researcher to name the various scales. This can often be done quite simply by examining the variables forming each scale. For example, from a perusal of the variables forming scale 1 (factor 1), we agree that both height and weight measure size of an individual, while scale 2 (factor 2) composed of occupation, education and income, measures social status. Thus we could name scale 1 as 'scale of size' and scale 2 as 'scale of social status'.

After the scales are constructed, the next step is to compute a particular respondent's score on a particular scale. Suppose our 5 items were chosen to measure the attitude of the respondents on the precondition of normal weight. We code the question from 1 (strongly disagree) to 5 (strongly agree). Thus if we have '5' against the variable 'health' we mean that the respondent 'strongly agreed' that 'height' is a factor in maintaining normal weight. Now if a respondent scored 1 on item 1 (height), 2 on item 2, 4 on item 3, 5 on item 4 and 1 on item 5 for scale 1, then his or her scale score for scale 1 is

$$0.58 (1) + 0.68 (2) + 0.55 (4) + 0.47 (5) + 0.65 (1) = 7.14$$

where 0.58, 0.68, ..., 0.65 are the factor loadings respectively for item 1, item 2, ..., item 5.

EXERCISES 8

1. What do you understand by the term 'measurement' in research? Define measurement error. Compare and contrast random error, systematic error and situational error with examples.
2. Define reliability and distinguish it from validity. Does reliability ensure validity? Explain the concept of reliability in the context of stability, equivalence and internal consistency.
3. What are the various techniques of measuring reliability? Distinguish test-

retest method from parallel forms method of measuring reliability. How is a reliability coefficient estimated using these methods? Elaborate.

4. How does the split-half method of measuring reliability differ from other methods? Describe the steps involved in estimating reliability using split-half method.
5. Discuss the utility of Spearman-Brown prophecy formula. Describe how this formula provides an estimate of the effect of shortening or lengthening a test on its reliability coefficient.
6. Describe Kuder-Richardson formula 20 and 21 for estimating reliability of a test. In what respects formula 20 differs from formula 21? The mean score of a 50-item test is 40 with a variance of 75. Estimate the reliability of this test.
7. How does ANOVA technique work in assessing internal consistency test reliability? Describe Cyril Hoyt method in this context.
8. What is standard error of measurement? How is the standard error related to reliability coefficient? If a person's observed score in a test is 100, with standard deviation 5 and reliability coefficient 0.80, what can be said about the person's true score with 95% certainty?
9. Suppose that a given test has reliability of 0.90 and a standard deviation of 9, find the standard error of measurement. Compute also the 95% confidence interval if a student were to get a score of 65 on the test.
10. Compare concurrent validity with predictive validity and construct validity. What are the steps involved on testing the construct validity?
11. What is scaling? Enumerate the importance of scaling in social science research. Distinguish between rating scales and ranking scales with examples. Compare and contrast simple category scales, simple response scales and multiple response scales.
12. What is Likert scale? How is scale score computed for each individual in such scaling? Describe the basic procedure for the construction of Likert scale. A researcher wishes to compare two hotels on the following attributes: (i) Convenience of location (ii) friendly personnel and (iii) value for money. (a) Design a Likert scale to accomplish the task (b) Design a semantic differential to accomplish the task, (c) Design a graphic rating scale to accomplish the task.
13. Compare and contrast the following in terms of their relative merits of and problems with:
 - (a) Rating and ranking scales
 - (b) Summated and cumulative scales
 - (c) Scalogram analysis and factor analysis
 - (d) Unidimensional and multidimensional scaling
14. (a) Describe in brief the procedure for developing a scalogram and illustrate the same by an example.
(b) Given the following information:

Number of subjects=50, Number of items=5 and Number of errors=15.
Compute the Guttman's coefficient of reproducibility.

15. A test expert wishes to find the whole-test reliabilities for the following split-half correlations, all of which were calculated on the basis of a sample of 20 persons. Find the whole-test reliability for each of the following split-half correlations:
(a) 0.83, (b) 0.76, (c) 0.92, (d) 0.50 and (e) 0.40
16. An expert wishes to discover how long a set of subsets must be to reach a reliability value of 0.90. For each subset, the following significant correlations were computed. Estimate how long each subset should be to obtain the 0.90 reliability value:
(a) 0.89, (b) 0.86, (c) 0.75, (d) 0.60 and (e) 0.70
17. For a given administration of the SAT verbal, the mean score was 500 with a standard deviation of 100. The test-retest reliability was 0.95. Obtain the standard error of measurement. If the reliability had been reduced to 0.80, find the standard error of measurement.
18. Using the standard errors calculated in Q. 18, find the 95% confidence intervals for the SAT verbal scores of (a) 300, (b) 400, (c) 500, and (d) 600.
19. The standard error of measurement on a certain test has been found to be 15.03 with a standard deviation of 50. Find the reliability coefficient for the test using Rulon formula.
20. A new intelligence 150-item test is being standardized, and it has been found to have a mean of 100 and a standard deviation of 15. Find its Kuder-Richardson 21 internal consistency reliability.
21. Write true (T) or false (F)
 - (a) The Spearman-Brown prophecy formula must always be used when assessing reliability via the test-retest technique.
 - (b) The Kuder-Richard formula 21 is used to determine internal consistency reliability.
 - (c) The standard error of measurement is related to reliability such that the higher the value of the standard error of measurement the lower the resulting reliability value.
 - (d) In general, increasing the number of test items typically increases the test's reliability.
 - (e) The reliability of any test depends on the value of the correlation, regardless of whether the correlation is significant or not.
 - (f) If a test is proved to be reliable, it must necessarily be valid.
22. Below is a Likert-type scale that might be used to evaluate your opinion of the educational program you are in. There are five response categories: strongly disagree through strongly agree, 1 being assigned for strongly disagree and being assigned to strongly disagree. The complete scoring scheme is follows: 1=Strongly disagree; 2: Disagree; 3: Neither agree nor disagree; 4: Agree; 5: Strongly agree.

A. The program is very challenging

1 2 3 4 5

B. The general level of teaching is good

1 2 3 4 5

C. I really think I am learning a lot from the program

1 2 3 4 5

D. Students' suggestions are given serious attention here

1 2 3 4 5

E. The program does a good job of preparing one for a career

1 2 3 4 5

F. The program is above my expectations.

1 2 3 4 5

(a) How would the different items be valued?

(b) Record your answers to the above items.

(c) In what two different ways could such responses be used?

(d) What would be the purpose of each?